Search Algorithms in Large Spatial Data using Analytical Digital Circle Definition and OpenMP Library

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Abstract. This paper deals with algorithms to search in large Spatial data using analytical digital circle definition in a regular grid, superimposed on spatial data to generate pixels or clusters that contain different Referenced sites. The nearest clusters and the nearest sites in clusters, as a response to the request from a position in the central cluster, are the results of analytical circle application. Algorithms to draw analytical digital circle in digital geometry lead to establish serial and parallel algorithms, that integrates OpenMP primitives to facilitate searching in large spatial data. Analysis shows integration of analytical digital circle definition.

Keywords: Large Spatial Data, GIS, Analytical Digital Circle

1 Introduction

Spatial data search are applications in eHealth, eMarketing, eCommerce to find the nearest sites with available services in a region. Spatial Indexation is a technique, used to improve data search in spatial database. Spatial indexation uses different tevhniques as grids, R-tree and R+tree : Kholladi in [13] proposed comparisons between these indexations and R-tree is an excellent technique to be implemented in a spatial database.

Spatial data emphasized localisation informations to describe objects as points, lines, polygons. Spatial data is defined by a tuple (id, data_1, data_2, ..., data_{n-1}, data_n) where id references directly or indirectly to GPS and different data_i describes all the data corresponding to GPS references id.

Many systems of coordinates and conversion between each other in GIS exist to reference to an object. For instance a function f((x_1,... x_i, ...x_n))=(y_1, ...y_i,... y_m) of coordinate conversion is still possible, to have (y_1, ...y_i,... y_m) as euclidian coordinates.

In the context of big spatial data, traditional methods to read data in spatial database are limeited. It appears important to integrate parallelism in data processing and the choice of Nosql dabatabas in order to improve the time taken by the database, to give a response. In this way, a framework map-reduce has been proposed by many scientists in data processsing. The problem is in GIS, according to an initial position, what are the nearest sites? How to determine the nearest site? According to specific fields, as pharmarcy sites, the question is how to determine the nearest sites from a fixed position? The question of the nearest sites can have many applications to Medical Centers, vehicule sellers. The question depends on the notion of neighborhood (4-neighborhood, 8-neighborhood), which also use distance definition. There are various methods to obtain the shortest road, the shortest path, as Dijkstra's algorithms, to determine the nearest object, as K-NN algorithms, k-means algorithm.

In digital Geometry, Antoine Vacavant presented in his thesis in [11] many variants of grid as regular grids and irregular grids. Bresenham, Reveillès and Andres introduced respectively definitions for digital lines and digital circles as Bresenham's line, Bresenham's circle, reveilles's analytical line, Eric Andres ' sanalytical line, Eric andres 's circle

Our motivation is concerned with introduction of digital object in Digital Geometry, precisely the definition of analytical digital circle as regular circle proposed by Eric Andres, to establish algorithms to search in large spatial data.

A regular grid is superimposed on GIS data before algorithm application to search spatial data. Combination of GIS and Digital Geometry are proposed to determine the nearest clusters and the nearest sites in clusters.

Parallelism contributes to improve data search in large spatial data, as the framework map-reduce used in [1], [3], [4], [6].

This paper presents parallelism algorithm based on openMP to facilitate data search.

Therefore, this paper is organized as follows : section 2 concerns with preliminaries. It presents different concepts related to quasi-affine application and analytical digital circle. Section 5 describes the method of spatial data search : It shows search algorithms based on an analytical search definition.

2 Preliminaries

There are many representations of spatial data, such as an euclidian representation, UTM, GPS. It also exists different conversions between each repere. The idea is to consider the representation of spatial data in a two dimensional space to consider a euclidian repere.

Each pixel corresponds to a cluster that contains spatial data, precisely point sites. This section recalls the definition of a regular grid and an irregular grid that will be superimposed on the surface of GIS data.

This section also focuses on the definition of analytical circle. There are many techniques in Digital Geometry for tiling a surface and to describe a model of digital objects.

2.1 Tiling a surface with a regular grid

A regular grid can be generated by a quasi-affine application defined by Nehlig and al. in [8] and also presented by Jacob et al. in [9]

Definition 1. *quasi-affine transformation [8]*
quasi-affine transformation is a tuple (w, M, V) $\in \mathbb{Z}xM_n(\mathbb{Z})x\mathbb{Z}^n$ (We consider that det(M) $\neq$ 0) defined by the following application :

$$\mathbb{R}^n \rightarrow \mathbb{R}^n$$
$$X \mapsto [\frac{MX+V}{w}].$$

$$\textit{where} \quad M = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} & m_{15} \\ m_{21} & m_{22} & m_{23} & m_{24} & m_{25} \\ m_{31} & m_{32} & m_{33} & m_{34} & m_{35} \\ m_{41} & m_{42} & m_{43} & m_{44} & m_{45} \\ m_{51} & m_{52} & m_{53} & m_{54} & m_{55} \end{bmatrix} \quad , \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \quad \textit{and} \quad V = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{bmatrix}$$

Figure 1 shows a quasi-affine application in a two dimensional space. A cluster is created in the intersection of straight lines.

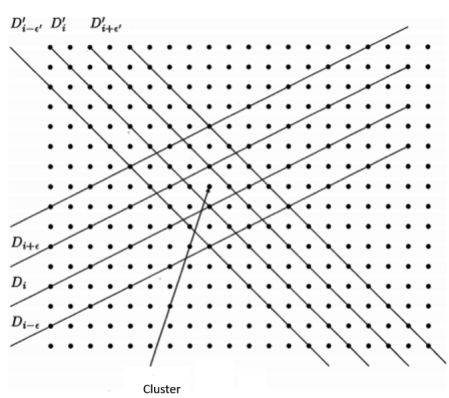


Fig. 1: Quasi affine application in a two dimensional space

Quasi affine application can easily generate an orthogonal repere in 2D or 3D space. Mesh generation [5] is important to split data to make easy parallelism application to divided data through clusters.

2.2 Analytical Digital Circle

In Digital Geometry, a set of shapes has been defined, such as analytical hypersphere in a n-dimensional space. In a two dimensional space, analytical hypersphere become analytical digital circle. According to Eric Andres et al. in [12] , analytical digital circle definition is proposed in the following definition :

Definition 2. *Let $O(x_o, y_o)$, r and ω be respectively the center, the radius and the thickness where $(x_o, y_o) \in Z^2$ and $(r, \omega) \in R^2$. Analytical digital circle is a set of points $(x, y) \in Z^2$ defined by : $(R-\frac{\omega}{2})^2 \leq (x_o - x)^2 + (y_o - y)^2) \leq (R+\frac{\omega}{2})^2$*

Analytical digital circle is illustrated in Figure 2.

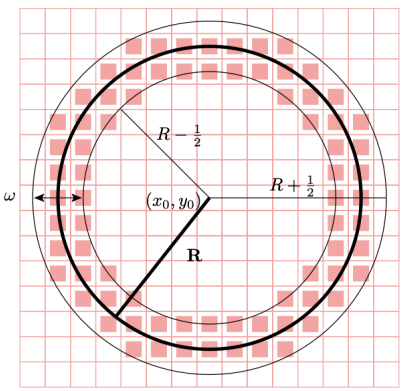


Fig. 2: analytical digital circle

Eric Andres et al. in [12] proposes several kinds of ananlytical digital circles as supercover circle, standard, naive circle That can be used to build search in spatial data. He propose an algorithm to draw a regular circle, a case of analytical circle : In our context, a pixel in a regular circle becomes a cluster.

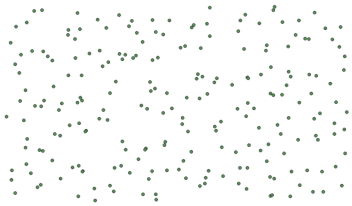
The notion of circle is important because it allows to take into account all the directions for a path leaving the initial position, represented by the center of the circle. Different levels of analytical digital circle have been considered, according to the radius going to 1 to n, if not success to find data.

3 Method Description

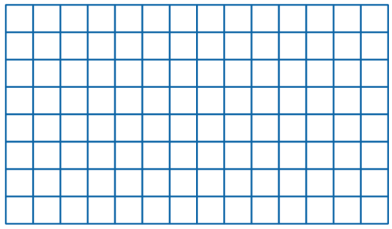
This section describes the applications of mesh and analytic digital circle to search in large spatial data. Mesh applications in image processing to represent pixels or geometric shapes in a grid are introduced in spatial data : neighborhood in a regular grid is applied to spatial data. search algorithms in spatial data with a regular grid are also proposed.

3.1 Mesh Generation

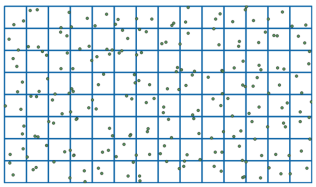
The same mesh generation proposed by sere and others in [7] is considered. It can be realized by a quasi affine transformation. For instance, figure 3a shows point sites that can be medical centers or sellers of cars. A regular grid in figure 3b is superimposed on point sites, to have the result proposed in figure 3c.



(a) point sites



(b) A regular grid



(c) A superimposition of a regular grid on point sites

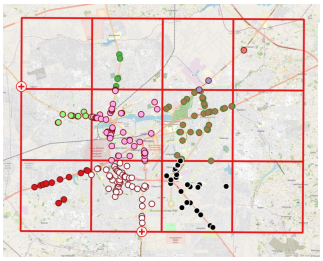


Fig. 4: A superimposition of a regular grid on a real map

Figure 4 also shows an example of supêrimposition on a real map

3.2 Search Algorithms in large spatial data

Drawing an analytical circle leads to determine clusters : if the centers of clusters are the members of the circle, then, all the sites in the cluster are concerned with the search. In following analytical digital circles of Eric Andres with regular circle definition, algorithm 1 presents a sequential search of data in clusters, represented by pixels in the regular circle.

Algorithm 1: Search of spatial data

```
Input: ;
ImageI: Image with the superimposed regular grid in order ;
data : data to search in the clusters;
r : the radius of the center;
x_center, y_center : the coordinates of the center where the request is sent.
;
Output: ;
Sites that contains data, in the clusters in the circle
Begin;
x ← 0;
y ← r;
d ← r − 1;
while (y ≥ x) do
    SearchingCluster(x_center + x, y_center + y, data, ImageI);
    SearchingCluster(x_center + y, y_center + x, data, ImageI);
    SearchingCluster(x_centre − x, y_centre + y, data, ImageI);
    SearchingCluster(x_center − y, y_center + x, data, ImageI);
    SearchingClusterr(x_center + x, y_center − y, data, ImageI);
    SearchingCluster(x_center + y, y_center − x, data, ImageI);
    SearchingCluster(x_center − x, y_center − y, data, ImageI);
    SearchingCluster(x_center − y, y_center − x, data, ImageI);
    if (d ≥ 2x) then
        d ← d − 2x − 1;
        x ← x + 1;
    else if (d < 2(r − y)) then
        d ← d + 2y − 1;
        y ← y − 1;
    else
        d ← d + 2(y − x − 1);
        y ← y − 1;
        x ← x + 1;
End
```

In the context of large spatial data, parallelism is useful to process all the data: algorithm 2 takes into account OpenMP library to facilitate data research : it also calls algorithm 3.

Algorithm 3 gives more details on spatial data search in a specific cluster through the function SearchingCluster : It uses several functions, as described :

- the function findClusteraddress (...) returns the cluster at (x, y). It takes as inputs the image, ImageI. The model represented in figure as the matrix is built through loading the image Image.
- the function findallsitesaddress (...) gives as a list, all the site references in the cluster.
- the function findrequest (...) verifies in the spatial database where indexation is made on site references, to find speedily if the data is in the sites in the list of sites.

A variant of algorithm 3 is proposed in algorithm 4 that searchs directly the list of sites in the spatial database where indexation is made on cluster references.

4 Implementation analysis and Discussions

Consider an image with the size 500 x 500. A matrix is superimposed on this image to obtain a grid.

The conception considers that the search of spatial data in following the virtual grid representing the matrix is realized through a program, implemented in c++. This program corresponds to serial or parallel algorithms 1 and 2 : they read the matrix, as a vector in c++. In C++, STL library proposes data structures and algorithms as the structure vector :

- At the first case , the matrix is represented by $vector < vector < vector < int >>>$ matrix, a structure in three dimensional space (3D). The matrix has columns in y-axis, rows in x-axis and height in Z-axis, as illustrated in figure 5. Algorithm 3 is used in this case to obtain references to point sites. The cell in the matrix contains Integer references that are associated to different sites in MongoDB. Indexation on site references facilitates searchs in spatial data.
- at the second case, the matrix is in two dimensional space : it corresponds to a $vector < vector < int >>$ matrix, illustrated in Figure 6. Algorithm 4 is used in this case to get point sites. Each cell of the matrix is in two dimensional space and contains one integer reference for a cluster in MonogoDB. Indexation on cluster references, improves spatial data search.

In all the cases, there is always a relation between clusters and sites in the spatial database.

Comparisons between algorithm 3 and Algorithm 4 are studied in indexation for mongoDB side and in algorithm side.

Algorithm 2: parallel search of spatial data

Input: ImageI: Image with the superimposed regular grid;
data : data to search in the clusters;
r : the radius of the circle;
x_{centre}, y_{centre} : the coordinates of the center where the request is sent;
;
Output: Sites that contains data, in the clusters in the circle
Begin;
$x \leftarrow 0$;
$y \leftarrow r$;
$d \leftarrow r - 1$;
while $(y \geq x)$ **do**
  $omp_set_num_threads(8)$;
  `#pragma omp sections` { ;
  `#pragma omp section` ;
  SearchingCluster($x_{centre} + x$, $y_{centre} + y$, data, ImageI);
  `#pragma omp section` ;
  SearchingCluster($x_{centre} + y$, $y_{centre} + x$, data, ImageI);
  `#pragma omp section` ;
  SearchingCluster($x_{centre} - x$, $y_{centre} + y$, data, ImageI);
  `#pragma omp section` ;
  SearchingCluster($x_{centre} - y$, $y_{centre} + x$, data, ImageI);
  `#pragma omp section` ;
  SearchingCluster($x_{centre} + x$, $y_{centre} - y$, data, ImageI);
  `#pragma omp section` ;
  SearchingCluster($x_{centre} + y$, $y_{centre} - x$, data, ImageI);
  `#pragma omp section` ;
  SearchingCluster($x_{centre} - x$, $y_{centre} - y$, data, ImageI);
  `#pragma omp section` ;
  SearchingCluster($x_{centre} - y$, $y_{centre} - x$, data, ImageI);
  } **if** $(d \geq 2x)$ **then**
    $d \leftarrow d - 2x - 1$;
    $x \leftarrow x + 1$;
  else if $(d < 2(r - y))$ **then**
    $d \leftarrow d + 2y - 1$;
    $y \leftarrow y - 1$;
  else
    $d \leftarrow d + 2(y - x - 1)$;
    $y \leftarrow y - 1$;
    $x \leftarrow x + 1$;
End

4.1 Difference between Algorithm 3 and Algorithm 4

In algorithm side, one of the difference between the two cases is that the first case gets the references of points sites in the matrix, in memory. That could improves

Algorithm 3: SearchingCluster1

Input: ImageI: Image with the superimposed regular grid;
data : data to search in the clusters;
x : x reference of the cluster ;
y: y reference of the cluster;
Cluster : a cluster;
liste : List of sites in the cluster;
listesitesfind: liste of sites that contains the data;
;
Output: liste of sites that contains data
Begin;
listesitesfind$\leftarrow null$;
cluster$\leftarrow findClusteraddress(x, y, ImageI)$;
liste$\leftarrow findallsitesaddress(cluster)$;
listesitesfind$\leftarrow findrequest(liste, data)$;
print (listesitesfind);
End

Algorithm 4: SearchingCluster2

Input: ImageI: Image with the superimposed regular grid;
data : data to search in the clusters;
x : x reference of the cluster ;
y: y reference of the cluster;
Cluster : a cluster;
listesitesfind: liste of sites that contains the data;
;
Output: liste of sites that contains data
Begin;
listesitesfind$\leftarrow null$;
cluster$\leftarrow findClusteraddress(x, y, ImageI)$;
listesitesfind$\leftarrow findrequest(cluster, data)$;
print (listesitesfind);
End

the time taken to find in the spatial database through indexation on clusters, proposed in the second case. Algorithm 3 contains more instructions than algorithm 4. For instance, the instruction liste$\leftarrow findallsitesaddress(cluster)$ in Algorithm 3 does not exists in Algorithm 4 : it consists of getting the references of point sites from the matrix in the program. The program can be executed in localhost in a latop or inside the spatial database in the server.

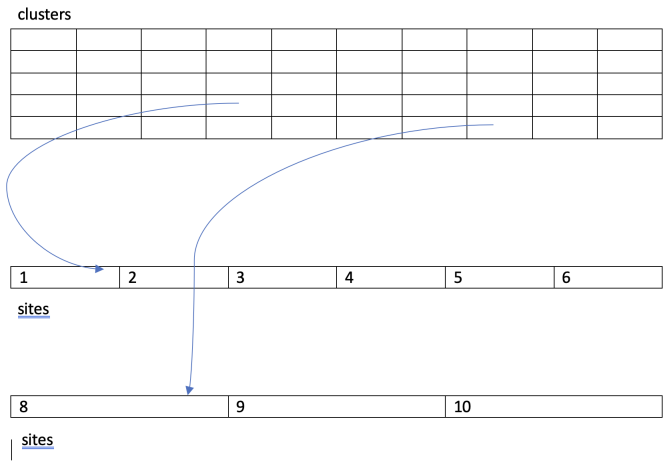


Fig. 5: first case : Matrix in 3D

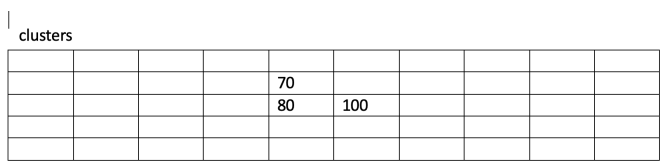


Fig. 6: second case : Matrix in 2D

4.2 Indexation on clusters and indexation on points sites in spatial MongoDB

Let us examine the contain of the function findrequest which is different in the two algorithms. In algorithm 3, the function connect to mongoDB and returns the list of point sites that contains the data, while in Algorithm 4, it verify directly in the spatial database the sites found.

Comparisons of indexation on clusters and indexation on point sites allows to determine or to estimate the time taken with the function findrequest in Algorithm 3 and Algorithm 4.

Object reference in 2D is simple for application in computing. It considers the distance, as the crow flies and does not take into account the path to go to the nearest sites : the length of the path (road) is not used. Obstacles can also occur in the road, even in path consideration.

5 Conclusion and perspectives

This paper shows the application of analytical circles, precisely, regular circles defined by Eric Andres in spatial data search. sequential and parallel algorithms

to search spatial data have been proposed in this way. Simulation consider a matrix with informations in cells.

Future researches will focus on analysis of analytical digital circles in an irregular grid as a grid obtained by kmeans followed by Voronoi diagram and integration in Qgis as a plugin to facilitate spatial data search.

The notion of distance is based on as the crow flies, as criteria because an obtacle can occurs.

Future works will also analyze other methods in artificial intelligence to take into account obtacles in processing the nearest sites. Framework map-reduce integration to replace OpenMP library with comparisons is also an option to be study in perspectives.

In digtal geometry, digital curves passing through pixels, can be explored in GIS, where pixels are clusters of sites and digital curves are the models of roads.

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